

Online Resource 1 (ESM_1)

Reciprocal-free divisors of prescribed order over finite fields and the saturation cost of cyclic quantum synchronizable codes

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This Online Resource collects material moved from the main article to meet the page limit. Section numbers, lemma numbers and reference labels ([A] = Li and Zhu, Int. J. Theor. Phys. 61, 251 (2022); [B] = Du et al., IEEE Access 8, 8449 (2020)) refer to the main article. The text is taken unchanged from the full version of the manuscript.

S1. Proof of Lemma 2.3 (dual containment)

Lemma 2.3 (dual containment). Let $\gcd(N, q) = 1$, and let C be a cyclic code of length N with generator polynomial g_C . Then

$$C^\perp \subseteq C \iff g_C(x) \mid g_{C^*}(x) \mid x^N - 1.$$

Proof.

Step 1 (orthogonality as a polynomial condition). For $u, v \in R_N$ put $\bar{v}(x) = \sum v_j x^{-j} \in R_N$. The coefficient of x^s in $u(x)\bar{v}(x)$ is $\sum_j u_{j+s} v_j$, with indices mod N . This is the inner product of v with the s -th cyclic shift of u . Since C is spanned by the cyclic shifts of g_C ,

$$v \in C^\perp \iff g_C(x) \bar{v}(x) = 0 \text{ in } R_N.$$

Step 2 (identify $C^\perp$). Write $x^N - 1 = g_C h_C$. Then $g_C \bar{v} = 0$ in R_N if and only if h_C divides $\bar{v}$ (as an element of R_N). The map $v \mapsto \bar{v}$ is the ring automorphism of R_N induced by $x \mapsto x^{-1}$. It carries $\langle h_C \rangle$ onto $\langle h_C(x^{-1}) \rangle = \langle h_{C^*} \rangle$, because $x^{\deg h_C}$ is a unit of R_N . Hence

$$C^\perp = \langle h_{C^*} \rangle.$$

Step 3 (compare ideals). Since h_{C^*} is a monic divisor of $x^N - 1$, $C^\perp \subseteq C$ holds if and only if $g_C \mid h_{C^*}$. Taking reciprocals, this is equivalent to $g_C \mid h_C$, and that is equivalent to $g_C g_{C^*} \mid g_C h_C = x^N - 1$. ■

S2. Class 2 computational observations (full version of Sect. 11.4)

Class 2 computational observations

For [A]'s BCH class, $n = (q^m - 1)/(q - 1)$ and $N = 2n$. The general index rule in [A] is only partially legible in the available source. The following observations therefore use a reconstructed rule:

- the defining-set truncations as printed in [A]’s Example 3;
- the ranges $1 \leq i \leq \delta - 2$ and $1 \leq j \leq \delta^* - 2$;
- [A]’s stated upper limits for δ and δ^* .

“Saturated” means $\text{ord}(h) = N$ under this reconstruction.

(q, m)	N	$\text{ord}_N(q)$ $= w_0 = w$	parameter tuples	saturated	$\text{ord}(h) < N$	range of s (saturated)
(3, 2)	80	4	75	75	0	8–24
(5, 2)	312	4	1,490	1,482	8	6–66
(7, 2)	800	4	10,269	10,263	6	8–124
(3, 3)	728	6	19,908	19,896	12	12–144

In every saturated tuple observed, $s > w$.

These are computational observations under a reconstructed rule. They are not a theorem, and they do not cover [A]’s Class 2 exhaustively. The tuples with $\text{ord}(h) < N$ may simply be excluded by [A]’s exact hypotheses. The observed minimum $s = 6$ for $(q, m) = (5, 2)$ is below $2w = 8$, so the bound $s \geq 2w$ of Theorem 11.1 is not claimed for Class 2.

S3. Computational verification record (full version of Sect. 12.3)

Computational verification record

Reference implementation.

- w and w_0 were computed by a dynamic programme over the lcm-cover form (Proposition 4.2).
- This programme was cross-checked against an independent brute force that enumerates sets of q -cyclotomic cosets directly. That brute force does not use Proposition 4.2.

Correction of the enumerator. The first implementation of the support-reduction enumerator for Theorem A reported 7 spurious failures (for example $(q, N) = (23, 225)$). The error was in the enumerator, not in the theorem: it allowed each prime support to be used **at most once**. Theorem A allows distinct blocks to share a support, as in §6.6, where 45 and 75 both have support $\{3, 5\}$. After correction, every test below reported zero failures.

Check	Range (finite)	Number of checks	Failures
DP vs independent coset brute force	15 values of q (even and odd), $N < 200$	1,999	0
Lemma 5.1	all prime powers $q < 128$; odd $m < 5000$	99,981	0
Theorem A	all prime powers $q < 128$; odd $N < 2000$	39,968	0
Formula (6.3) for w_0	as for Theorem A	39,968	0
Theorem B	all prime powers $q < 128$; odd two-prime $N < 20000$	195,405	0

Check	Range (finite)	Number of checks	Failures
Theorem C(i)	all prime powers $q < 128$; odd primes $p < r < 400$	30,079	0
Theorem C(ii)	explicit triples (p, r, q) with $p, r < 400$	594	0
Theorem 11.1 (i) and the factor-degree structure	odd $q < 128$; primes $n < 400$	1,081	0

In total about 409,000 checks were run. The ranges were chosen deliberately to include adversarial cases:

- repeated prime powers;
- three and four distinct primes;
- mixed 2-adic classes;
- blocks requiring helper primes;
- supports used by more than one block;
- even q , for Lemma 5.1 and Theorems A and B.

The record is not an exhaustive verification over all inputs.

[A] instances. These were recomputed directly: generator degrees, $\text{ord}(h)$, dual containment and exact $d(D)$. In Example 3, the factor $m_0 = x - 1$ printed in $[A]$'s g_1 and g_3 is incompatible with dual containment and with the printed dimensions. This was recorded explicitly (§11.3) rather than corrected silently.

S4. Code

The Python scripts are provided as Online Resource 2 (ESM_2.zip); see its README.